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**Six year efficacy of silvicultural treatments to control American beech
regeneration in stands affected by beech bark disease in Ontario, Canada**

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Abstract

High beech regeneration density is a concern in northern shade tolerant hardwood forests. High densities of beech (*Fagus grandifolia* Ehrh) regeneration can crowd out other desirable species, such as sugar maple (*Acer saccharum* Marsh.), and jeopardize long-term sustainability since beech is under threat from beech bark disease (*Cryptococcus fagisuga*/*Neonectria* spp. Complex). We examined the efficacy of three tending (no tending, brush saw, and basal bark herbicide) and two timing and harvesting (deferred 5 years post-single tree selection harvest, concurrent with uniform shelterwood harvest) treatments on reducing beech regeneration density and promoting sugar maple regeneration density over 6 years. Six years after tending we found that large beech regeneration density was reduced, medium beech regeneration density had recovered to pre-tending levels in most treatments, and small beech regeneration density remained unaffected. Tending treatments had no effect on any size class of sugar maple regeneration density but the uniform shelterwood harvest promoted medium sugar maple density more than the single-tree selection harvest. Despite this increase, in all treatment combinations sugar maple regeneration densities remained below stocking targets. Our results suggest that while tending treatments can temporarily reduce beech regeneration densities, sugar maple is unable to take advantage of the increased growing space.

Keywords: beech bark disease, beech regeneration, competition control, single-tree selection, sugar maple regeneration, tolerant hardwood management, uniform shelterwood, vegetation management alternatives

1. Introduction

Beech bark disease (BBD) results from an insect-fungal disease complex. The invasive scale insect that initiates the disease complex, *Cryptococcus fagisuga* Lind, was introduced into North America in the late nineteenth century and has since slowly spread across Canada and the United States ([Cale et al. 2017](#)). The scale infestation is followed by a *Neonectria* fungal invasion that kills mature beech ([Houston 1994](#)). In Ontario, beech scale was first reported in 1966 ([Keizer 1988](#)), beech mortality from BBD was reported in 1981 ([Bisessar et al. 1985](#)), and *Neonectria* sp. was confirmed in 1988 ([McLaughlin and Greifenhagen 2012](#)). Since then, BBD has spread progressively across Ontario, with beech mortality rates as high as 99% in stands throughout much of the affected range ([McLaughlin and Greifenhagen 2012](#)). Given the impending loss of overstory beech, removal of beech regeneration to help regenerate shade tolerant hardwoods to a more sustainable condition is desirable.

American beech (*Fagus grandifolia* Ehrh) occupies a range of ecological conditions and is widely distributed across the eastern part of North America: as far inland as eastern Texas in the south and central Ontario in the north. Beech has unique features such as extreme shade tolerance and longevity, prolific seed production, and aggressive root sprouting ([Tubbs and Houston 1990](#)). Once established in a stand, American beech will persist and can withstand disturbances. Beech produces abundant mast in the form of beechnuts, a food source for a variety of species including black bears, marten, deer, raccoons, squirrels, and blue jays ([McCullough et al. 2001](#)). The loss of mature beech trees from BBD may have long-term effects on wildlife species that use beech for food and shelter, and has been shown to reduce carbon storage in these forests ([Kish et al. 2022](#)).

In the managed forests of central Ontario, about 624,000 ha of stands in the Great Lakes-St. Lawrence (GLSL) forest region have $\geq 10\%$ of American beech (Figure S1). In Ontario, American beech is generally found in association with sugar maple (*Acer saccharum* Marsh). Beech and sugar maple mixtures are traditionally managed using a single-tree selection harvest that typically removes up to a third of the stand basal area (BA) with re-entry every 20–30 years (OMNRF 2015). Selection harvesting promotes diameter growth amongst high-quality residual trees and releases suppressed understory trees (OMNR 2004). In Ontario, single-tree selection is focused on sugar maple regeneration but it also creates ideal conditions for beech regeneration, as beech grows well under low light conditions (Bohn and Nyland 2003; Nolet et al. 2008). The positive effect of single-tree selection on beech regeneration is compounded by the recent lack of sugar maple regeneration in these systems. Reasons for reduced sugar maple regeneration in absolute terms, and relative to beech, are varied. Sugar maple struggles to regenerate on high pH and low calcium soils in absolute terms and relative to beech (Cleavitt et al. 2021; Ouimet et al. 2017) suggesting that acid deposition may be a driving factor in beech proliferation in northern shade tolerant hardwood forests (Duchesne and Ouimet 2009). Other drivers of reduced sugar maple regeneration compared to beech are preferential browse by deer (Bédard et al. 2022; Henry et al. 2021), impacts of earthworms (Hale et al. 2006), phytotoxicity of beech (Hane et al. 2003), and lack of establishment or being outcompeted by beech in single-tree selection (Bédard et al. 2022; Dumont et al. 2023; Henry and Walters 2023; Leduc et al. 2024; Nolet et al. 2008; Rogers et al. 2021). The persistent rise of beech regeneration in comparison to sugar maple poses a serious threat to sugar maple dominated forest productivity.

Given concerns about the impact of BBD on forest productivity, stands with $\geq 20\%$ beech abundance in Crown forests of Ontario are now often managed using a shelterwood system or

group openings that preferentially remove beech instead of single-tree selection. Although beech are being managed for removal, it has proven difficult to decrease the amount of beech regeneration because they are prolific root sprouters after harvesting, particularly in stands with a high beech component ([Bohn and Nyland 2003](#); [Houston 2001](#); [Nyland et al. 2006](#)). Beech sprouts generally occur when the roots are injured, causing adventitious buds to form, although sprouts can occur where an injury is not evident ([Tubbs and Houston 1990](#)). When mature beech stems die or are cut, the roots sucker and the stumps sprout to produce fast-growing advanced regeneration that is highly competitive and forms very dense thickets in the understory of harvested stands. These thickets perpetuate in the understory and typically outcompete other desirable species, such as sugar maple, yellow birch (*Betula alleghaniensis* Britt.), and northern red oak (*Quercus rubra* L.) ([Giencke et al. 2014](#)). Affected trees and root sprouts that arise from susceptible rootstocks have been identified as potential hosts for new BBD infections on understory trees as small as 5 cm in diameter ([Cale et al. 2017](#); [Giencke et al. 2014](#)). These susceptible rootstocks often produce a secondary killing front in aftermath forests before regenerating trees reach the upper forest canopy ([Cale et al. 2017](#); [Nyland 2008](#)). While overstory-only interventions (i.e., preferential removal of beech during harvest) are enticing, their efficacy may be limited especially when beech advanced reproduction or overstory presence is already high ([Leduc et al. 2024](#); [St-Jean et al. 2021](#)). Evidence indicates that beech has positive density dependence for seedlings but negative density dependence for saplings ([Ramage and Mangana 2017](#)). So, it is possible that removing many overstory beech will release saplings from negative density dependence, resulting in a flush of new beech saplings in place of the removed overstory beech. If vegetation management is not applied in these forests, a cycle of

103 beech regeneration followed by eventual infection and death will substantially reduce the
104 productivity of a stand and lead to permanent low-quality beech.

105 Without treatment, beech thickets can affect not only future species composition, but also
106 stand growth and yield ([Hane et al. 2003](#); [Nyland et al. 2006](#)). Both mechanical and chemical
107 controls are potentially effective in reducing understory beech thickets depending on the size and
108 structure of the beech and the canopy openness ([Nyland et al. 2006](#)). Mist blown triclopyr and
109 glyphosate ([Ostrofsky and McCormack Jr 1986](#)), backpack application of glyphosate ([Bose et al.](#)
110 [2018](#)), and "hack and squirt" injections of glyphosate and basal spraying of triclopyr are also
111 highly effective for controlling beech understories ([Kochenderfer et al. 2004](#)). However, unlike
112 targeted applications with backpack and basal bark sprayers, mist blown herbicides are not
113 targeted and may damage species other than beech. Mechanical treatments such as site
114 preparation using a brush saw have also been suggested as an effective approach to control beech
115 thickets ([Nyland et al. 2006](#)). However, recent evidence suggests that mechanical removal of
116 beech advanced reproduction through brush saw and site preparation using a skidder with a blade
117 rake can cause vigorous resprouting of beech that can outcompete sugar maple and yellow birch
118 regeneration even after 11 years ([Dumont et al. 2023](#)); although yellow birch can outcompete
119 beech when fencing is used to remove browsing pressure ([Bédard et al. 2022](#)).

120 To address the emerging concern of overabundant beech regeneration in Ontario, we
121 used plot networks from two experimental sites established to examine the response of beech
122 regeneration to three tending treatments: control (i.e., untended), basal bark herbicide application
123 targeting beech between 1 and 20 cm in basal diameter, and motor-manual brush saw targeting
124 all beech less than 20 cm in basal diameter and two harvesting treatments: single tree selection
125 and uniform shelterwood. One site was harvested using single-tree selection in 2011 and the

tending treatments were applied in 2016 (deferred tending). At the second site, a regeneration cut following the uniform shelterwood system was applied in 2016, with tending treatments applied in the same year (concurrent tending). We sought to answer the following questions: i) Which tending treatments (mechanical, chemical) best control beech regeneration density?; ii) Was the efficacy of the tending treatments affected by timing or type of harvest?; iii) Did sugar maple regeneration respond to tending treatment or timing?; and, iv) Did initial abundance of beech overstory or regeneration affect the efficacy of the tending treatments?

2. Material and Methods

2.1 Study sites

Field trials were initiated in 2016 at two study sites in the Great Lakes-St. Lawrence Forest region of Ontario, Canada (Figure S1). The hardwood forest in this region is dominated by sugar maple, beech, and yellow birch. The soil at the study sites is typically dry to fresh sandy, low in nutrient and moisture availability, mostly non-calcareous, and contains a variable depth of coarse morainal deposits (Lee and Section 1998). The average annual temperature is 5.8 °C and the mean annual precipitation is about 1065 mm per year based on data from the nearest weather stations (https://climate.weather.gc.ca/climate_normals/index_e.html#1981).

At both sites we implemented three tending treatments: mechanical (brush saw, targeting all beech smaller than 20 cm in basal diameter with chainsaw when necessary), chemical (basal bark application of triclopyr at a concentration of 144 g L⁻¹, targeting beech 1 to 20 cm in basal diameter), and control (no tending). Tending treatments were replicated across the study areas in 4 ha (200 x 200 m) treatment blocks. Blocks were established to minimize within-block variation and maximize between-block variation. However, due to operational constraints, treatment areas were larger than the block areas and some blocks were clumped into the same treatment areas. At

the deferred tending site, tending treatments were applied in 2016, 5 years after a single-tree selection harvest. At the concurrent tending site, tending treatments were applied in 2017 during a uniform shelterwood regeneration harvest. The deferred tending site had 15 blocks, with 75 plots, and the concurrent tending site had 8 harvested blocks, with 40 plots (Figure 1). Basal bark treatments were reapplied in 2020 at the deferred tending site because many stems were missed or under-treated in 2016.

The deferred tending site was harvested using a single tree selection system in 2011. Although beech scale and disease were present at the time, beech trees were not specifically targeted for removal during the harvest. Rather, the objective was to improve quality by removing unacceptable growing stock, including diseased beech, as a general objective of single tree selection harvesting ([OMNR 2004](#)). After harvest and before tending (2016), the basal area of trees >8 cm in diameter at breast height (DBH) at our study plots averaged 18.3 m² ha⁻¹ and the average species composition by basal area (BA) was 54% sugar maple, 33% American beech, 4% yellow birch, and a combination of 15 other species representing less than 3% each.

At the concurrent tending site, the pre-tending, pre-harvest mean BA of trees >8 cm DBH was 23.5 m² ha⁻¹, composed of 65% sugar maple, 15% American beech, 6% red oak, 4% red maple (*Acer rubrum* L.), and a combination of 13 other species representing less than 3% each. The site was harvested in winter 2016, with a small portion completed in winter 2017, via a uniform shelterwood regeneration cut. Basal area, of trees >8 cm DBH, post-harvesting and tending was 16.8 m² ha⁻¹, composed of 67% sugar maple, 11% American beech, 7% red oak, 2% red maple, and 13 other species representing less than 2% each.

2.2 Data collection

Five permanent sample plots were established in each tending block (Figure 1). Permanent sample plots consisted of one 1000 m² large overstory (trees ≥ 17 cm DBH) subplot, one 400 m² small overstory (trees 8–16.9 cm DBH) subplot, and eight 16 m² regeneration subplots (trees < 8 cm DBH). In the regeneration plots, large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm) regeneration were assessed in 16 m², 8 m², and 4 m² nested plots, respectively. Regeneration plots were distributed in the permanent sample plots at the four cardinal directions at both 10- and 15 m from plot centre (Figure S2). Shoots from a common stem were assessed as separate stems if they emerged below the ground or at the root collar, or where the main stem was cut; otherwise, they were assessed collectively as a single stem. Seed-origin beech regeneration and root sprouts were not differentiated. At the sixth year remeasurement, visual assessment was used to differentiate beech sprouts regenerating from the cut face of brush saw stumps from all other beech regeneration. In the overstory plots, a health code was assigned to all stems in the plot based on crown health indicators following [Millers \(1991\)](#). Briefly, health was assessed according to six codes. Healthy (1) with less than 10 percent dieback, light decline (2) with between 10 and 25 percent dieback, moderate decline (3) with between 26 and 50 percent dieback, severe decline (4) with more than 50 percent dieback while still being alive, dead - natural (5), and dead – human caused (6).

All plots were measured pre-tending at both the deferred and concurrent tending sites. For the concurrent tending site, pre-tending measurements were also pre-harvest. Plots were remeasured in years one, three, and six post-tending at the deferred tending site and at years one and six post-tending at the concurrent tending site.

2.4 Data analysis

2.4.1 Tending effects on beech regeneration

We examined the response of large, medium, and small beech and sugar maple regeneration density to tending treatments through a generalized linear mixed effect repeated measures model with a negative binomial distribution and a log link function. Each regeneration size and species were fit separately. The model is described as:

$$\log RD_{ijklm} = \mu + H_i + T_j + Y_k + HT_{ij} + HY_{ik} + TY_{ik} + HTY_{ijk} + \pi_l + \rho_m \quad (1)$$

where RD_{ijk} is the regeneration density (total count of regeneration in a given size class in all eight nested subplots in a given overstory plot) representing the i th timing treatment and the j th tending treatment at the k th year of measurement in the l th plot and m th tending block, H_i is the fixed effect of harvest type and tending timing, T_j is the fixed effect of tending treatment, Y_k is the fixed effect of measurement year relative to when tending occurred, $\pi_{(l)}$ is the random effect of plot, and $\rho_{(m)}$ is the random effect of block. Models were fitted in R statistical software, version 4.1.0 ([R Core Development Team 2021](#)) using the *glmmTMB* package ([Brooks et al. 2017](#)). We used the same model structure to analyze the basal area weighted average of health condition of beech in each overstory plot through time. However, we used a linear mixed effect repeated measures model with a normal distribution. Model assumptions were visually confirmed using diagnostic plots. For all models we used the *emmeans* package to perform *post-hoc* analyses to determine significant differences between treatments ([Lenth et al. 2018](#)).

2.4.2 Initial abundance of beech effect on treatment efficacy at year six

We modified Eq. (1) by including the pre-treatment beech overstory and pre-treatment small beech regeneration density to test the relationship between initial beech abundance and treatment

efficacy. Initial beech abundance was estimated as basal area of live overstory beech and abundance of beech small regeneration (since some treatments did not remove it) in the pre-treatment plot. We used the *dredge* function from the *MuMIn* package ([Barton and Barton 2015](#)) in R statistical software to select the most parsimonious model based on Akaike Information Criterion (AIC) selection. All possible combinations including all three-way interactions were used in model selection allowing us to examine which variables were the most important in explaining beech and sugar maple regeneration densities (Table S1 & S2).

3. Results

3.1 Tending effects on beech regeneration

The response of beech regeneration density to the tending treatments was not significantly different between the deferred tending and concurrent tending site (Table 1, small regeneration $P=0.124$, medium regeneration $P=0.619$, large regeneration $P=0.682$). Both brush saw and basal bark tending initially reduced small and medium beech regeneration but, by the sixth year post-tending, small and medium beech regeneration densities had increased to pre-tending levels (Figure 2). An exception was the medium beech regeneration in the basal bark treatment plots at the deferred tending site, which continued to decline between the first and sixth year post-tending. Large beech regeneration densities did not spike, remaining significantly lower than pre-tending levels in both brush saw and basal bark tending treatments six years after tending (Figure 2). Six years post-tending about 40% of small and 25% of medium beech regeneration in brush saw treatments clearly regenerated by sprouting from cuts at the concurrent site (Table 2). Although about a third of large beech regeneration were sprouts from brush saw cuts at the concurrent tending site (Table 2), large regeneration density amounted to only three stems across all brush saw plots. Six years post-tending about half of small and a third of medium beech

regeneration in brush saw treatments clearly regenerated by sprouting from cuts at the deferred tending site, while there was no large regeneration that had clearly sprouted from cuts (Table 2). In control plots, medium and large beech regeneration densities increased slightly through time at the concurrent tending site but were relatively stable at the deferred tending site (Figure 2).

3.2 Tending effects on sugar maple regeneration

Regeneration density of small sugar maple significantly increased in the six years following harvest at the concurrent tending site regardless of treatment but declined significantly at the deferred tending site in the control and basal bark treatments (Figure 2). Medium maple regeneration density increased in all treatments at the concurrent tending site but did not change significantly through time at the deferred tending site in any treatment despite a decreasing trend in the control (Figure 2). Large maple regeneration density trended downwards in both the control and basal bark treatments, but the trend was only significant for the brush saw treatments at the concurrent tending site (Figure 2). Large maple regeneration densities did not differ significantly through time at the deferred tending site despite a downward trend at the control site (Figure 2).

3.3 Overstory dynamics

Live beech basal area was largely removed at the concurrent tending site and decreased slightly at the deferred tending site (Figure S4). The average basal area of beech before tending was 3.0 m² ha⁻¹ at the concurrent tending site but was 6.1 m² ha⁻¹ at the deferred tending site. Tree health codes of beech were generally stable at the concurrent tending site with a slight decrease (i.e., increase in crown condition) through time. The decrease was only significant in brush saw treatments (Figure 3). Tree health codes increased (i.e., decrease in crown condition) significantly in overstory beech at the deferred tending site across all treatments (Table S3,

Figure 3). Before tending, beech regeneration density was generally positively related to live beech overstory density in all size classes (Figure S5). Sugar maple basal area generally increased in all treatments and sites between the first- and sixth year post-tending with no differences between tending treatments (Figure S4).

3.4 Initial regeneration abundance effects on tending efficacy

Initial density of small beech regeneration was a significant predictor of the presence of medium beech regeneration six years post-tending (Table S4, Figure 4, $P < 0.001$). As the amount of pre-treatment small beech regeneration increased, the amount of medium beech regeneration at the sixth year also increased across all tending and timing combinations. Model selection omitted initial beech basal area when including initial small beech regeneration abundance. However, both tending treatment and timing remained important predictors of medium beech regeneration independent of small beech regeneration abundance. Six years post-tending, pre-treatment small sugar maple regeneration density did not affect medium sugar maple regeneration density in any of the timing or tending combinations (Table S4, Figure 5). Model selection omitted both initial beech basal area and initial sugar maple basal area from the explanatory predictors indicating that initial overstory abundance did not significantly affect sugar maple abundance at the sixth year post-tending.

4. Discussion

4.1. Tending effects on beech regeneration

Compared to pre-tending densities, tending reduced both medium and large beech regeneration in the first year. Small beech regeneration was largely unaffected because in most cases it was too small to be tended. By the sixth year post-tending, medium beech regeneration had recovered to pre-tending levels in most treatment combinations. Potential sources of medium beech in year

six are medium beech that survived tending, recruitment from small beech, or beech not present immediately post-tending but that recruited and developed to the medium regeneration class.

One mechanism for such “new” regeneration is resprouting from stumps, which contributed to between a third and half of the regrowth in brush saw plots. Previous work has suggested that these resprouts are unlikely to develop into vigorous overstory trees in the long-term ([Nyland et al. 2006](#)) but in the short-term they reduce growing space for the regeneration of other species. Resprouting following brush saw is a common occurrence for beech understory, allowing it to recover and dominate the understory within a decade ([Dumont et al. 2023](#)).

Basal bark applications increased beech medium regeneration at our concurrent tending site but decreased it at the deferred tending site. Note that the deferred tending site was re-treated with basal bark triclopyr 3 years after the initial application to remove missed stems and those where the treatment was misapplied. This application was the first broad scale use of triclopyr on public land in Ontario and contractors initially applied the herbicide to only half the base of the stem. This approach was not effective and subsequent applications (including at our concurrent tending site) were around the entire base of the stem. Whether beech medium regeneration at the deferred tending site recovers by year six, after the second round of basal bark treatment, remains to be seen. Given the comparable levels of post-tending small beech regeneration density in the basal bark and brush saw treatments at the deferred tending site, it seems likely that beech medium regeneration density will recover. Basal bark applications do not result in resprouting from a cut face but medium regeneration densities in basal bark treatments at the concurrent tending site still recovered to pre-tending levels by the sixth year. This finding, combined with our results that the pre-tending amount of small beech regeneration is an important driver of medium beech regeneration six years post-tending indicates that, regardless

of tending treatment used, medium beech regeneration will recover within six years if small beech regeneration is not addressed. Presence of small beech regeneration could be exacerbated following mast years ([Cleavitt and Fahey 2017](#)). Although beech seedlings do not typically grow as quickly or have as high survivorship as beech root sprouts ([Beaudet and Messier 2008](#); [Farahat and Lechowicz 2013](#)) they are an important component of beech regeneration due to their abundance ([Cleavitt et al. 2008](#)). Recovery in the large regeneration class was not evident but that is likely because six years is insufficient time for beech regeneration to recruit to this size class.

4.2 Tending effects on maple regeneration

While the immediate effect of the tending treatments was to reduce beech regeneration densities, the goal was to increase sugar maple regeneration densities. However, evidence did not indicate that any of the tending treatments increased sugar maple densities, despite that they increased growing space in the medium and large size classes. Even in the deferred tending basal bark treatment, where medium and large beech regeneration densities were below pre-tending levels at the sixth year, sugar maple medium and large regeneration densities did not increase. Analysis of basal area of maple large regeneration showed no significant increase through time suggesting that large sugar maple did not increase in size in the six years following tending treatments. It is possible at our deferred tending site that shade from the overstory was sufficient to keep sugar maple regeneration growth rates low. Crown closure recovers quickly following selection cutting, typically within 11 years ([Beaudet et al. 2004](#)). It is possible that the sugar maple regeneration was not being limited primarily by competition with beech regeneration but rather by limited access to light. Medium sugar maple regeneration did appear to respond to harvesting at the concurrent tending site, where it increased in all treatments. Increasing maple regeneration

density with higher harvest intensity has been documented in previous studies in northern temperate hardwoods ([Nolet et al. 2008](#); [St-Jean et al. 2021](#)) although recent evidence suggests that beech saplings can continue to outgrow sugar maple saplings if harvest is the only treatment applied ([Leduc et al. 2024](#)).

Even at our concurrent tending site where harvesting removed more basal area, sugar maple regeneration densities did not meet stocking targets. Stocking targets for sugar maple are elusive ([Henry et al. 2021](#)) but, for even aged harvest systems, targets proposed are 2500 stems per hectare of saplings 50–137 cm tall and 1000 stems per hectare of saplings >137 cm tall and <5 cm DBH ([Elenitsky et al. 2020](#)). For uneven aged systems, based on the q-value required for Ontario *sensu* [Arbogast \(1957\)](#), stems between 2 and 8 cm DBH should represent 450 stems per hectare ([OMNR 2004](#)) immediately post-harvest. In all cases, even where maple medium regeneration increased, on our sites stocking was below these criteria at the sixth-year post-tending. These findings align with a pattern of challenges with sugar maple regeneration in northern temperate hardwoods. Reasons are often region specific: impacts of earthworms ([Hale et al. 2006](#)), acid deposition increasing acidity and depleting base cations([Cleavitt et al. 2021](#); [Duchesne and Ouimet 2009](#); [Ouimet et al. 2017](#)), summer droughts ([Boakye et al. 2023](#)), browsing pressure ([Bédard et al. 2022](#); [Henry et al. 2021](#)), and beech phytotoxicity ([Hane et al. 2003](#)) have all been proposed. More research is needed to better understand the factors limiting sugar maple regeneration at our sites. Similar to findings of studies focused on mechanical control of beech, we did not find evidence that sugar maple regeneration could outcompete beech regeneration in either tending treatment ([Dumont et al. 2023](#)).

Our tending treatments were focused on medium and large beech regeneration; it is possible that pairing these tending treatments with a site preparation treatment (either mechanical

or chemical) soon after harvest could reduce small beech regeneration and allow sugar maple to both regenerate and benefit from the growing space afforded by the tending treatments. Mechanical site preparation may promote beech suckering from root injuries and would still need to be combined with follow up tending ([Dumont et al. 2023](#)). Chemical site preparation can also reduce regeneration of species other than beech although certain mixtures of herbicide and surfactants can promote sugar maple regeneration over beech regeneration ([Nelson and Wagner 2011](#)). Finally, it is also possible that sugar maple regeneration is limited by the acidity of the soils at the site ([Cleavitt et al. 2021](#); [Duchesne and Ouimet 2009](#); [Ouimet et al. 2017](#)) given that sugar maple regeneration densities were similar between tended blocks and control blocks. Soil amelioration through liming has shown promise to increase sugar maple growth rates in absolute terms and in comparison to beech ([Long et al. 2022](#); [Moore and Ouimet 2021](#)). Combining liming to ameliorate soil conditions and mechanical scarification to reduce beech regeneration has shown short-term promise in improving sugar maple and yellow birch regeneration compared to beech ([Bognounou et al. 2023](#)). Future studies should consider examining how soil amelioration, site preparation, and tending treatments could interact to improve sugar maple regeneration compared to beech regeneration in acidic soil types.

4.3 Overstory dynamics

At the deferred tending site, which had more beech overstory post-tending than the concurrent tending site, beech health decreased. Harvest intensity was higher (more stems were removed) at the concurrent tending site and, when possible, was focused on removing beech, so changes in average health condition were less pronounced. We also did not tag and follow stems through time, so stems moving from dead standing to laying on the forest floor would decrease standing basal area and result in a relative increase in health condition. Similar to the pattern of

deterioration at our deferred tending site, previous studies in the same forest region have shown that beech mortality is high and rapid following beech bark disease colonization ([Kish et al. 2022](#)). Despite the declining health condition of the beech overstory at the deferred tending site, in the untended plots no clear increase in beech regeneration that might be associated with beech thickets from root suckering as a response to the spread of beech bark disease was evident. Initial densities of beech were much higher than maple suggesting that, similar to other sites at the northern edge of beech range ([Nolet et al. 2008](#); [Roy and Nolet 2018](#)), beech regeneration density was already substantial at our sites before they were colonized by beech bark disease. Beech regeneration increases at the concurrent tending site were likely attributable to a suckering response caused by mechanical stimulation from harvesting operations ([Bohn and Nyland 2003](#); [Houston 2001](#); [Nyland et al. 2006](#)) and not a response to disease since health conditions at the site were stable due to limited beech presence after harvesting.

4.4 Initial regeneration abundance effects on treatment efficacy

Similar to previous studies, we found some indirect evidence of a positive density dependence between mature beech and small saplings ([Ramage and Mangana 2017](#)): before tending, as overstory beech basal area increased, small beech regeneration abundance also increased. However, unlike previous studies, our indirect evidence also showed a positive density dependence between mature beech and larger saplings ([Ramage and Mangana 2017](#); [Zhu et al. 2015](#)). In general, more overstory beech is indicative of more understory beech at our sites. While model selection indicated that initial beech regeneration density was a better predictor of medium regeneration density at the sixth year post-tending, initial beech overstory basal area was a significant predictor when initial beech regeneration density was omitted from the model. This finding suggests that at some level of live beech overstory density, beech regeneration density

will be too high to allow for effective control of the beech understory. Whether this relationship is driven by positive density dependence or increasing light availability due to high levels of beech mortality remains unclear.

Landscape-scale management of beech thickets is a challenge for forest managers in Ontario due to the high treatment costs. Given the effort needed to control beech thicket development, managing beech regeneration via tending may only be feasible for relatively small areas ([Giencke et al. 2014](#)). Our finding of lower beech regeneration density in plots with lower residual beech basal area and the positive (if not sufficient) effect of higher harvest intensities suggests that forest managers may consider preferentially removing beech from the overstory during harvest to stimulate sugar maple regeneration. However, future studies need to represent a broader range of overstory beech abundance and spatially map overstory tree abundance to properly determine thresholds of beech abundance that limit tending efficacy.

4.5 Conclusions

In conclusion, we found that both basal bark and brush saw tending effectively controlled large beech regeneration after six years post-tending, but medium regeneration density generally recovered to pre-treatment levels. This recovery was especially rapid when initial small beech regeneration was high. Tending treatments had no effect on sugar maple regeneration despite effectively releasing growing space in the regeneration layers; however, more intense harvesting may promote increases in sugar maple regeneration. Unknown is whether site preparation, which would clear microsites of small beech, combined with tending, which theoretically releases growing space for existing sugar maple, would be more effective. Despite increases in sugar maple regeneration at higher harvest intensities, stands in our study remained consistently understocked. Given the similar sugar maple regeneration densities in our treated blocks and

control blocks, our results suggest that sugar maple regeneration is being limited by factors other than competition from medium and large beech regeneration. Research is needed to better understand what is limiting sugar maple regeneration in this region, including the details of regeneration dynamics such as germination, recruitment, and stagnation.

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Author contribution statement

Searle, Eric (Corresponding Author): Formal analysis, Funding acquisition, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing.

Jones, Trevor A.: Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Yietagesu, Aklilu**

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Writing – review & editing. **Trerise, Bridget:** Writing – review & editing. **Bein, Aliya S.:** Writing – review & editing.

Competing interests

The authors declare there are no competing interests.

444 **Data availability statement**

445 Data and code required to reproduce the analyses in this paper are available at:

446 [10.6084/m9.figshare.25315291](https://doi.org/10.6084/m9.figshare.25315291)

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Tables

Table 1. Effect of tending treatment, timing of tending, and measurement year on regeneration density of beech and sugar maple in a central Ontario study. Each species and regeneration size class were fit with separate models. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm).

Predictors	Beech						
		Small		Medium		Large	
	df	χ^2	P	χ^2	P	χ^2	P
Tending	2	4.01	0.134	5.26	0.072	4.51	0.105
Timing	1	6.87	0.008	0.08	0.774	0.88	0.350
Year	2	47.6	<0.001	68.0	<0.001	99.5	<0.001
Tending:Timing	2	4.18	0.124	0.96	0.619	0.76	0.682
Tending:Year	4	22.8	<0.001	101	<0.001	264	<0.001
Timing:Year	2	55.0	<0.001	43.3	<0.001	15.8	<0.001
Tending:Timing:Year	4	2.82	0.588	7.62	0.106	35.3	<0.001

	Sugar maple						
		Small		Medium		Large	
	df	χ^2	P	χ^2	P	χ^2	P
Tending	2	6.33	0.042	10.7	0.100	0.49	0.782
Timing	1	3.36	0.067	19.6	<0.001	10.2	0.002
Year	2	10.0	0.007	27.2	<0.001	18.2	<0.001
Tending:Timing	2	0.17	0.920	3.32	0.345	0.29	0.864
Tending:Year	4	17.8	0.001	17.1	0.004	3.17	0.530
Timing:Year	2	49.2	<0.001	73.7	<0.001	14.8	<0.001
Tending:Timing:Year	4	5.29	0.258	5.18	0.269	9.63	0.047

Table 2. Regeneration density of beech by size class, tending timing, and whether the stem regenerated from a brush saw cut or another means, and total number of stems, in a central Ontario study. Number shows average stem density in each timing and size category, with standard error of the mean in parentheses. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm).

Site	Regeneration size class	Non-induced regeneration (number ha ⁻¹)	Sprout from brush saw cut (number ha ⁻¹)	Total regeneration (number ha ⁻¹)
Concurrent tending	Small	3406 (803)	2469 (934)	5875 (1657)
	Medium	3594 (533)	1297 (219)	4890 (708)
	Large	85.9 (31.8)	23.4 (23.4)	109.4 (48.3)
Deferred tending	Small	24792 (2876)	21938 (3128)	46730 (5874)
	Medium	5854 (524)	2594 (304)	8448 (817)
	Large	99.0 (16.8)	0 (0)	99.0 (16.8)

Figure captions

Figure 1. Treatment plot layouts for the beech bark disease silvicultural study at the deferred tending (a) and the concurrent tending (b) study sites in central Ontario.

Figure 2. Response of beech (a) and sugar maple (b) regeneration (regen) density to tending treatment, timing, and time since application in a central Ontario study. Filled points represent treatment means with lines indicating associated 95% confidence intervals. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm). A version of this figure that includes observed values is available as Figure S3.

Figure 3: Average health condition of standing beech by harvest and tending treatment through time in a central Ontario study. Background points show the basal area-weighted average health condition of standing beech for each overstory plot. Larger circles show the average basal area-weighted average health code for each year, tending, and harvesting treatment with associated 95% confidence bars. Health codes vary from 1 (no crown dieback) to 5 (dead, natural causes) and 6 (dead, human causes).

Figure 4. Relationship between initial beech small regeneration and change in sixth year post-tending beech medium regeneration, by tending and harvest treatment, in a central Ontario study. Solid lines are predicted model fits with associated 95% confidence bands subtracted from the pre-treatment average for the harvesting and tending combinations. Background points are observations of medium regeneration at the sixth year post-tending subtracted from the average pre-tending medium regeneration by tending and timing combination. Dashed line indicates when sixth year medium regeneration densities were equal to pre-tending densities. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm).

630 **Figure 5. Relationship between initial sugar maple small regeneration and change in sixth**
631 **year post-treatment sugar maple medium regeneration, by tending and harvest treatment,**
632 **in a central Ontario study.** Solid lines are predicted model fits with associated 95% confidence
633 bands subtracted from the pre-treatment average for the harvesting and tending combinations.
634 Background points are observations of medium regeneration at the sixth year post-tending
635 subtracted from the average pre-tending medium regeneration by tending and timing
636 combination. Dashed line indicates when sixth year medium regeneration densities equalled pre-
637 tending densities. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and
638 small (height 20–129 cm).

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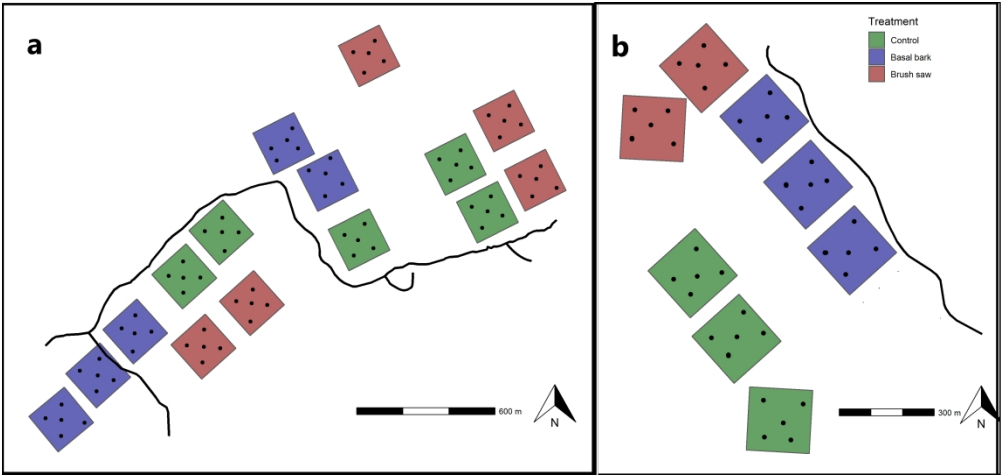


Figure 1. Treatment plot layouts for the beech bark disease silvicultural study at the deferred tending (a) and the concurrent tending (b) study sites in central Ontario.

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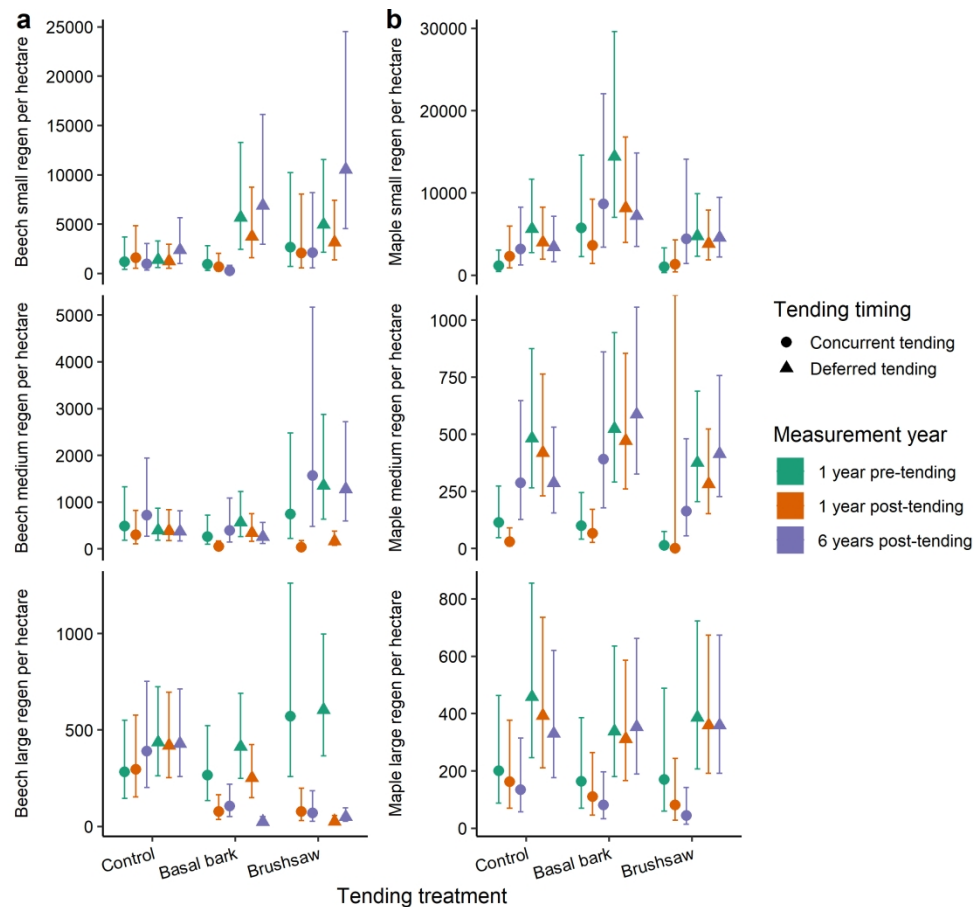


Figure 2. Response of beech (a) and sugar maple (b) regeneration (regen) density to tending treatment, timing, and time since application in a central Ontario study. Filled points represent treatment means with lines indicating associated 95% confidence intervals. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm). A version of this figure that includes observed values is available as Figure S3.

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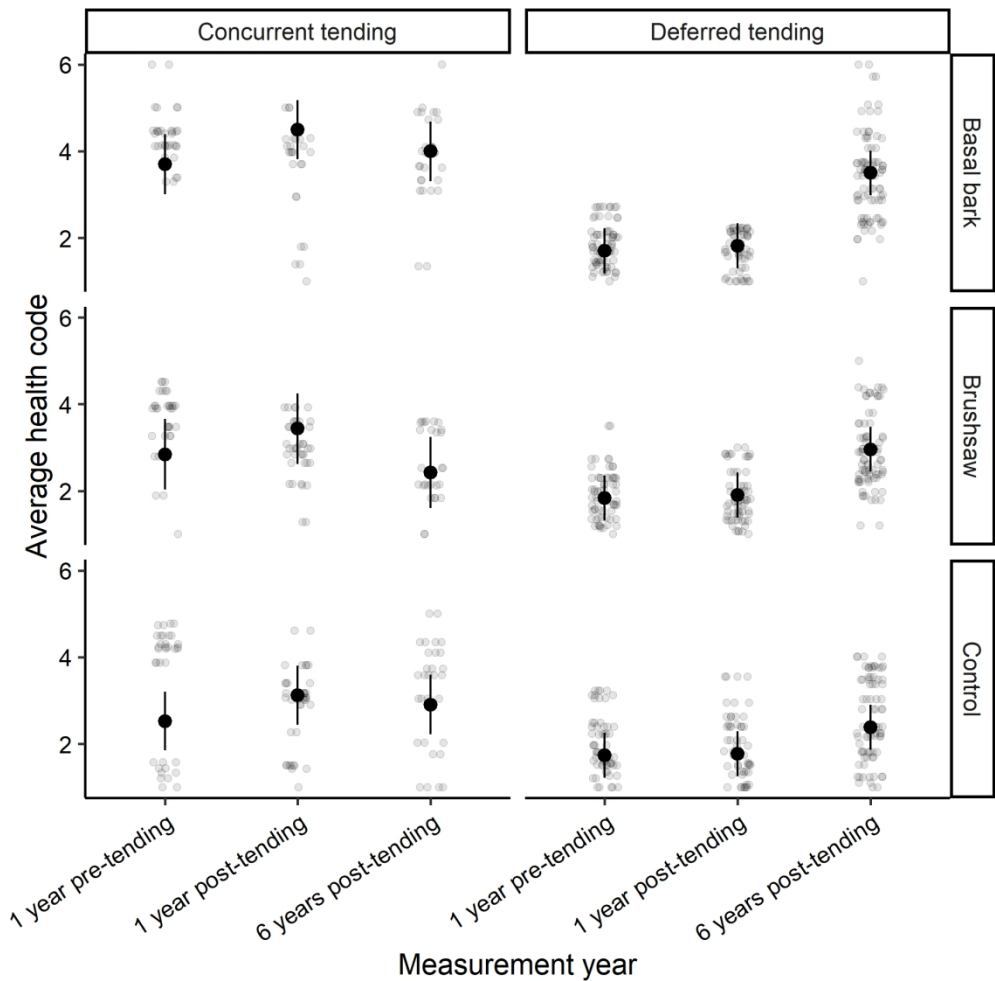


Figure 3: Average health condition of standing beech by harvest and tending treatment through time in a central Ontario study. Background points show the basal area-weighted average health condition of standing beech for each overstory plot. Larger circles show the average basal area-weighted average health code for each year, tending, and harvesting treatment with associated 95% confidence bars. Health codes vary from 1 (no crown dieback) to 5 (dead, natural causes) and 6 (dead, human causes).

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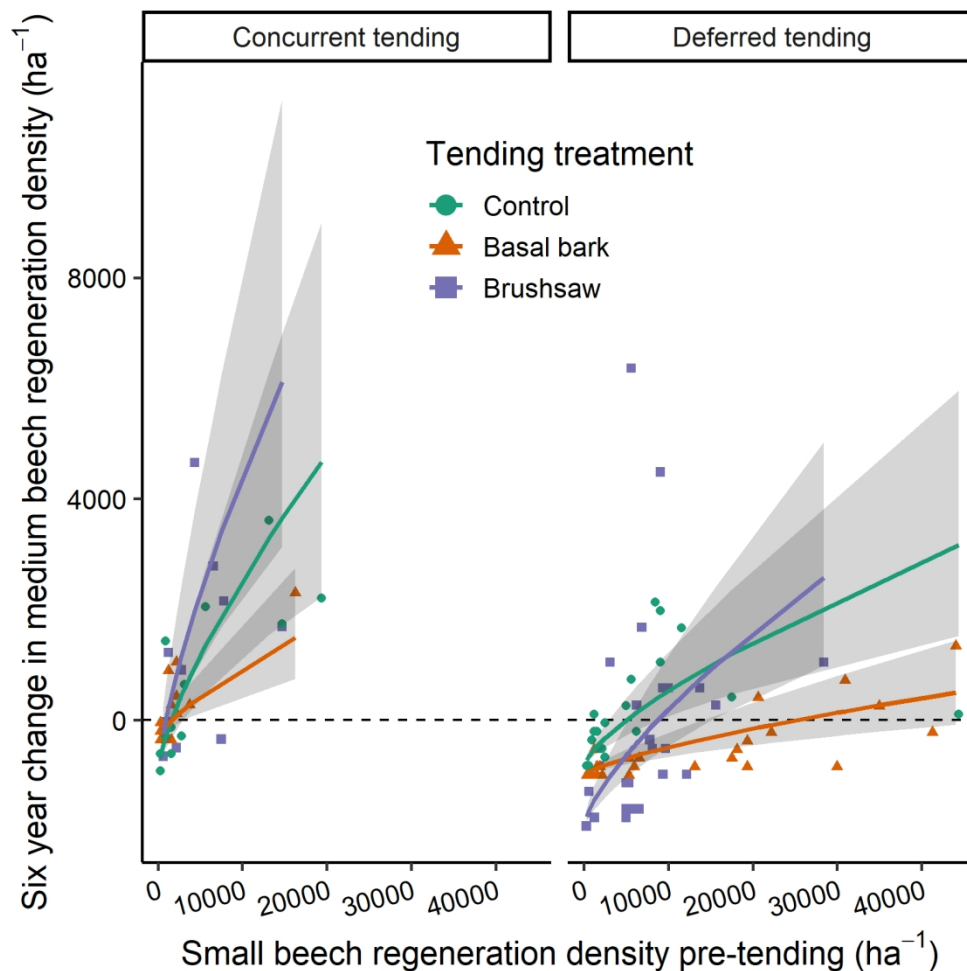


Figure 4. Relationship between initial beech small regeneration and change in sixth year post-tending beech medium regeneration, by tending and harvest treatment, in a central Ontario study. Solid lines are predicted model fits with associated 95% confidence bands subtracted from the pre-treatment average for the harvesting and tending combinations. Background points are observations of medium regeneration at the sixth year post-tending subtracted from the average pre-tending medium regeneration by tending and timing combination. Dashed line indicates when sixth year medium regeneration densities were equal to pre-tending densities. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm).

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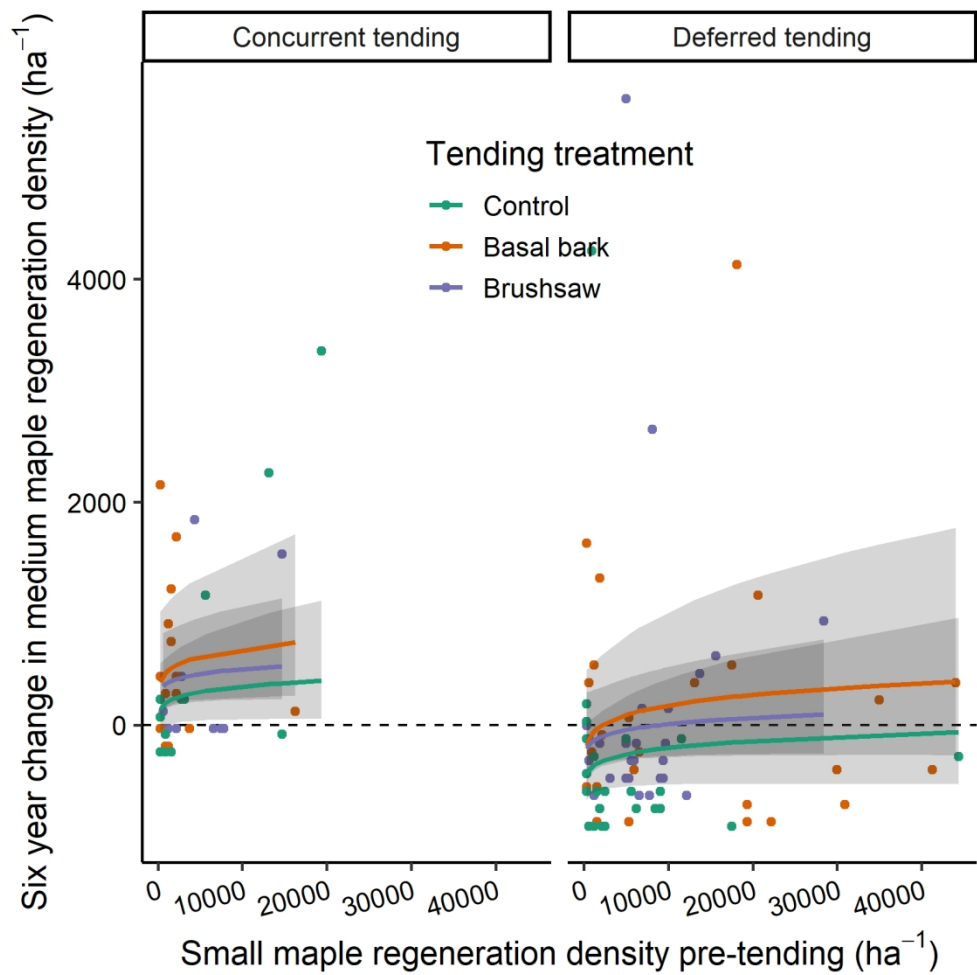


Figure 5. Relationship between initial sugar maple small regeneration and change in sixth year post-treatment sugar maple medium regeneration, by tending and harvest treatment, in a central Ontario study. Solid lines are predicted model fits with associated 95% confidence bands subtracted from the pre-treatment average for the harvesting and tending combinations. Background points are observations of medium regeneration at the sixth year post-tending subtracted from the average pre-tending medium regeneration by tending and timing combination. Dashed line indicates when sixth year medium regeneration densities equalled pre-tending densities. Size class refers to: large (DBH 2.0–7.9 cm), medium (DBH 0.1–1.9 cm), and small (height 20–129 cm).

148x148mm (300 x 300 DPI)